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REVIEW

Overview of WiFi fingerprinting-based indoor positioning

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Abstract
Location-based services have attracted significant attention since the concept of Industry 4.0 was proposed and the 'Internet +' era began owing to its social and commercial value. Many scholars have attempted to introduce machine learning in indoor fingerprint positioning to improve indoor positioning accuracy, enhance system robustness, reduce costs, and improve the performance of indoor positioning methods. A comprehensive overview of indoor positioning technology, its methods, and classifications is provided. Furthermore, a detailed review of the application of WiFi fingerprinting and machine learning methods in indoor positioning is presented, and the advantages and disadvantages of these methods when applied to indoor positioning are analyzed. This study summarizes the difficulties and challenges encountered by indoor positioning and suggests development directions.

1 | INTRODUCTION

The accuracy of satellite-based positioning has been demonstrated in the outdoor environment. This creates significant convenience in our daily lives, work, and other scenarios such as navigating cars and route planning. Numerous high-quality positioning software programs, such as Baidu Map and AMap, have been developed based on the global positioning system (GPS). These software programs are widely used in every aspect of society. In indoor environments, however, the obstruction of indoor objects and the absorption effect from the walls will significantly attenuate the satellite signal during outdoor to indoor transmission. Consequently, the satellite signal is usually very weak inside a building; hence, it is difficult to achieve precise positioning of indoor objects. Thus, new technologies are required to enhance indoor positioning. Indoor positioning is more challenging compared with outdoor positioning owing to the complex nature of indoor environments. The following factors must be considered: (1) the presence of physical obstacles, (2) signal fluctuation and noise, (3) wall blockage, (4) multiple available signal propagation paths, (5) non-line-of-sight distance (LoS) between two points, (6) movement of personnel and objects, (7) temperature variation, and (8) interference from electrical signals. Therefore, indoor positioning requires higher robustness and positioning accuracy.

The main contributions of this study are as follows:

1) This work introduces the techniques used for indoor localization and compares these techniques in terms of accuracy range, advantages, and disadvantages.

2) This study presents a review of the applications of WiFi fingerprinting and machine learning methods to indoor positioning, and an analysis of the advantages and disadvantages of these methods in indoor positioning research.

The rest of this study is structured as follows:

In Section 2, the main indoor positioning technologies are discussed. In Section 3, the main WiFi-based indoor positioning methods are presented, including the range-based WiFi indoor positioning method and the WiFi fingerprint-based indoor positioning method. Section 4 presents a review of the applications of WiFi fingerprinting and machine learning methods in indoor positioning, and an analysis of the advantages and disadvantages of these methods applied to indoor positioning research. Finally, Section 5 presents the main conclusions of the study.

2 | INDOOR POSITIONING TECHNOLOGY

Because humans spend most of their time indoors, it is of great importance to explore indoor positioning technology. Several different wireless signals have been used by researchers

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to realize indoor positioning. Some mainstream indoor positioning technologies include ultrasonic indoor positioning, radio frequency identification (RFID) indoor positioning, ultra-wideband (UWB) indoor positioning, Bluetooth indoor positioning, infrared indoor positioning, ZigBee indoor positioning, and WiFi indoor positioning.

2.1 | Ultrasonic

Ultrasound is a sound wave with a frequency higher than 20,000 Hz, and the ultrasonic positioning system consists of electronic tags with fixed positions and multiple mobile units (MUs). Knowing the speed of sound, the distance between each mobile unit and the fixed electronic tag can be obtained by calculating the travel time from the transmitter to the receiver, that is, the time of flight (ToF). The position of the target to be located is determined by the geometric relationship between the mobile unit and the fixed reference point [1]. The deployment difficulty is low, and the positioning accuracy is up to the cm level. However, one of the characteristics of sound wave, namely, the transmission speed of the sound wave, is significantly affected by the air density, and the air density is mainly affected by the humidity. Therefore, the accuracy of this positioning method is significantly affected by the climate. Positioning has a small effective range and is susceptible to obstacles, whereas accurate distance measurement requires tight synchronization between the transmitter and receiver's local clocks.

2.2 | RFID

Radio frequency identification technology mainly uses radio frequency integrated circuits for two-way data communication of electromagnetic wave signals. RFID positioning systems are generally composed of RFID readers and electronic tags, and the two communicate with each other via a fixed protocol. When an RFID reader placed on an object to be positioned detects multiple tags with known coordinates in a given reference frame, the position of the object to be positioned can be estimated using a sensor fusion-based dynamic estimation algorithm [2]. Low-cost and easy-to-deploy, passive tag-based RFID systems are widely used to locate tagged items in indoor scenarios such as warehouses, libraries, and stores [3]. This technology has a short operating distance, generally a maximum of tens of meters, and is not easy to integrate into other systems.

2.3 | Ultra-wideband

Ultra-broadband technology is considered the most emerging wireless communication technology in recent years, and is quite different from conventional communication technologies [4]. UWB positioning systems use prearranged anchor nodes and bridge nodes with known positions to communicate with newly added blind nodes, and use triangulation or 'fingerprint' positioning to determine the position. This technology eliminates path loss and multipath effect interference, and the positioning accuracy is high. The limiting factor is that the ultra-wideband positioning device is expensive, which limits its application in a wide range of scenarios.

2.4 | Bluetooth

The most commonly used Bluetooth positioning technology is the low-power Bluetooth technology based on Bluetooth 4.0, that is, iBeacon technology, which has low power consumption, faster connection speed, high transmission rate, stable and safe signal transmission, and no interference. In the indoor positioning system based on low-power Bluetooth technology, the Bluetooth beacon is first laid in the positioning area, and the beacon continuously broadcasts signals and data packets to the surrounding. The RSSI value is for the caculation of the specific location using the built-in positionning algorithm of the mobile phone. The advantage of Bluetooth positioning is that it is simple to implement, and the positioning accuracy is closely related to the laying density and transmission power of Bluetooth beacons. The error is still high owing to dense multipath fading that often occurs in the indoor environment [5]. With the release of Bluetooth 5.1, the Bluetooth protocol has added new functionalities such as direction finding, which makes the application of Bluetooth low energy (BLE) technology more advantageous.

2.5 | Infrared

The wavelength of infrared rays used in positioning systems is generally in the range of 830–950 nm, which is an electromagnetic wave with a wavelength between radio waves and visible light waves. The infrared positioning system locates the object and attaches an electronic tag that emits infrared rays; it then measures the distance or angle of the signal source using multiple infrared sensors placed in the room, thereby calculating the location of the object [6]. The requirements for the positioning environment are high, and the positioning effect is relatively good in an environment with relatively few occlusions. Heat sources and lights will have a certain effect on infrared rays.

2.6 | Zigbee

Zigbee is a short-range, low-power communication protocol based on the IEEE802.15.4 standard [7]. The ZigBee indoor positioning solution technology forms a network between several blind nodes to be located, which are reference nodes whose positions are known, and gateways. The gateway receives information from the above nodes and sends it to the positioning program in the server to realize positioning. The advantage is that information exchange between nodes and nodes and gateways requires only a small amount of energy, resulting in low power consumption and low price. The disadvantages include a

small positioning range, large positioning error, large multipath effect, and poor anti-interference capability.

2.7 | WiFi

WiFi is a wireless local area network based on the IEEE802.11b standard. WiFi networks are widely used in various indoor environments such as homes, hotels, shopping malls, offices, campuses, and hospitals. Most smartphones, laptops, and other portable user devices currently support WiFi. The objective of WiFi indoor positioning is to achieve positioning detection and tracking tasks in complex environments using a wireless local area network composed of wireless access points (including wireless routers). It uses a combination of empirical tests and signal propagation models to locate the connected mobile devices based on the location information of network nodes. The RSSI-based fingerprint positioning method is the current mainstream WiFi positioning method [8], and the positioning accuracy depends on the calibration, that is, density of points.

2.8 | Hybrid

Any isolated positioning technology has its uniqueness and limitations. The advantages and disadvantages of various single-beacon positioning technologies can be explored and combined so that they learn from each other, improve positioning accuracy, and reduce model complexity. For instance, in [9], UWB technology and WiFi technology were used for positioning. The advantages of ultra-wideband stability and high-precision positioning combined with the advantages of WiFi, without the need to introduce specific infrastructure, improved the positioning accuracy of the positioning system. The study in [10] combined UWB technology and BLE technology for localization. The system not only leverages the stability and high precision performance of ultra-wideband signals, but also the advantages of BLE technology for easy deployment and low power consumption. In [11], the researchers reported that BLE provides a higher accuracy than WiFi, but only at a low application range, whereas Wi-Fi can be used in a wider application range; combining BLE and Wi-Fi networks can improve the overall accuracy and scope of the system.

The advantages, disadvantages, and positioning accuracies of these techniques are summarized in Table 1.

Among these technologies, the ultrasonic, RFID, UWB, Bluetooth, infrared, and ZigBee indoor positioning techniques require specific hardware support and several positioning beacons, such as infrared, ultrasonic, or RFID beacons. WiFi is the most widely deployed indoor wireless infrastructure. The WiFi positioning technique does not require additional hardware. It can provide indoor positioning by solely using the evenly distributed wireless network that is set up in the building and the smart device at the user's end. Because of the increasing popularity of using WiFi, WiFi signals are now ubiquitous in the society as they meet the infrastructure requirement of WiFi positioning. In addition, many people have become habituated

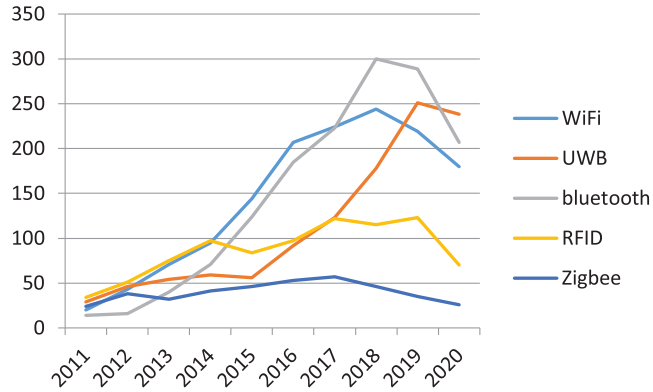

FIGURE 1 Publication trends: main techniques in indoor positioning

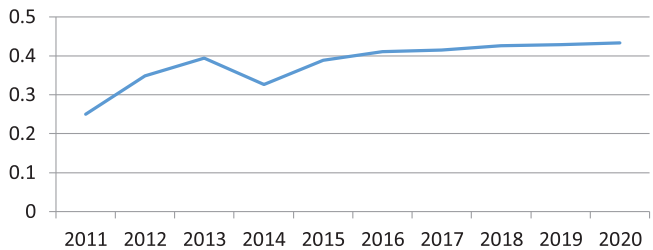

FIGURE 2 Proportion of WiFi indoor positioning studies focusing on the WiFi fingerprint technique

to using WiFi. Consequently, WiFi positioning has a substantial advantage over other types of indoor positioning techniques. Therefore, this study focused on the review of WiFi fingerprint indoor positioning technology.

To better understand the level of research interest in WiFi-based indoor positioning, we analyzed research records covering the past decade (2011–2020) from the web of science databases using techniques such as keyword searches. Because the number of publications identified using different keywords varied significantly, we performed a normalization process. Table 2 presents the acquired data, and the publication trends are illustrated in Figures 1 and 2. After analyzing the data, we observed that WiFi is one of the top three popular indoor positioning techniques. Subsequently, we analyzed its advantages compared with UWB and Bluetooth. In addition, we found only one relevant review paper on WiFi-based indoor positioning technique [19]; however, the paper considered active and passive positioning and focused on signal transmission.

This study focused on the WiFi fingerprint technique, which accounted for almost 50% of all the reviewed publications on the WiFi-based indoor positioning technique, as shown in Figure 2. However, there is no relevant review paper on this technology.

3 | INDOOR POSITIONING METHODS

So far, almost all WiFi indoor positioning systems are based on the received signal strength (RSS). In general, these systems can

TABLE 1 Major indoor positioning technologies

Positioning technique	Advantage	Disadvantage	Positioning accuracy	System
Ultrasonic indoor positioning [1]	High positioning accuracy	Multipath effect, thermal drift effect, high cost, severe decay	Centimeter scale	Bat [8]
RFID indoor positioning [2]	Low cost, high speed	Short distance, low communication capacity	Centimeter scale	LANDMARC [9]
Ultra-wideband (UWB) indoor positioning [3]	Strong resistance to interference, strong penetration, high positioning accuracy	High cost	Sub-meter scale	Dart UWB [10]
Bluetooth indoor positioning [4]	Low power consumption, easily deployable	Short distance	Meter scale	BIPS [11]
Infrared indoor positioning [5]	Low cost, mature technology	Poor resistance to interference	Meter scale	Active badge [12]
Zigbee indoor positioning [6]	Low cost, low power consumption	Poor stability	Meter scale	[13]
WiFi indoor positioning [7]	No additional hardware required for deployment, low cost, wide application range	Tedious fingerprint collection, degeneration of WiFi signal	Meter scale	RADAR [14]

TABLE 2 Recent publications about indoor positioning

Years \ Keywords	WiFi	UWB	Bluetooth	RFID	Zigbee	WiFi fingerprint
2011	20	29	14	34	24	5
2012	43	46	16	51	38	15
2013	71	54	40	75	32	28
2014	95	59	71	97	41	31
2015	144	56	124	84	46	56
2016	207	92	185	97	53	85
2017	224	123	223	122	57	93
2018	244	178	300	115	46	104
2019	219	251	289	123	35	94
2020	180	238	207	70	26	78

be classified into two categories: (1) indoor positioning based on the measurement of the distance (range-based method) and (2) indoor positioning based on a WiFi fingerprint.

3.1 | Range-based WiFi indoor positioning method

The range-based indoor positioning method relies on the measurement of the distances between transceivers using the time of arrival (TOA), time difference of arrival (TDOA), and angle of arrival (AOA). In these methods, the RSS is first converted to a distance that uses the signal propagation model. Subsequently, the location of the mobile client is calculated using the geometric method (e.g. trilateral measurement). The positioning principles and their respective advantages and disadvantages are summarized in Table 3.

The deployment of TOA, TDOA, and AOA-based positioning techniques is tedious and expensive because they require certain hardware or software modifications on the access point (AP). In particular, a set of angle measurement sensors or high-precision timers must be used to measure the AOA or TOA, respectively. These devices can increase the cost of the positioning system. However, the distance between the AP and the user is calculated using the RSS propagation model. The decay of each AP signal is determined by the travelling distance, as well as several environmental factors, such as human, wall, and humidity. Therefore, it is impossible to obtain a general signal propagation model that can appropriately simulate the real-world condition. In reality, the propagation of the wireless signal in a complex indoor environment is subject to multiple interferences, including the multipath effect, reflection, refraction, and diffraction. The positioning obtained based on non-LoS propagation often has

TABLE 3 Major range-based indoor positioning methods

Positioning method	Principle	Limitation	Advantage
Based on the time of arrival (TOA) [16]	Calculate the distance between the transceivers by measuring the time taken for the WiFi signal to travel from the emitter to the receiver. This distance can be used to provide the positioning	Clock synchronization, line-of-sight propagation	Simple algorithm
Based on the time difference of arrival (TDOA) [17]	Calculate the difference of the distance between the transceivers by measuring the time difference taken for the WiFi signal to travel from the emitter to the receiver. This distance difference will then be used to provide positioning	Line-of-sight propagation	Simple algorithm
Based on the angle of arrival (AOA) [18]	Conduct positioning using a triangle shape obtained using the angle–side–angle rule	Prone to interference from the environment	Simple algorithm

a larger error, and fails to provide a satisfactory positioning performance.

3.2 | WiFi fingerprinting-based indoor positioning

The RADAR positioning system is the first task in WiFi fingerprint-based indoor positioning. WiFi fingerprinting (a process of signal collection and association with indoor locations) is a promising approach, without assuming LoS. In this method, the characteristic of the location is associated with the mode of the detected signal (e.g. the RSS vector emitted from different WiFi APs). Therefore, when the exact location of the AP is unknown, fingerprint recognition can provide positioning information without requiring distance or angle measurement data. This feature ensures that the positioning accuracy is not affected by multipath effect or non-LoS propagation. Therefore, the WiFi fingerprint-based positioning method is highly feasible for indoor positioning. In this study, the WiFi fingerprint-based positioning method is considered as it offers significant advantages compared with other indoor positioning methods.

3.2.1 | Positioning principle

Wireless APs are first deployed in the building and are labelled as AP1, AP2, … , APn. By scanning the AP signals at the reference point in the environment with a smart terminal, the signals that are emitted from different APs and the corresponding MAC addresses that are associated with these APs can be obtained. The intensity of the received signals is represented by the RSS.

During propagation, the WiFi signal will experience a path loss. Specifically, the power density of the signal decreases with an increase in the distance that follows the path loss model. The intensity of the WiFi signal emitted from the same AP becomes unique at different travelling distances. Each location is represented by a unique RSS vector (RSSi1, RSSi2, … , RSSin) based on the signals emitted from multiple APs. This one-to-one relationship between the RSS vector and the location is similar to the concept of a human fingerprint, and can be used as a unique identifier of the location information. Therefore, the RSS vector measured at a specific location can be used to predict its actual geological position.

3.2.2 | Positioning process

The WiFi fingerprint-based positioning process can be classified into two steps [23]: offline data collection stage and online matching stage.

Step 1: It is impossible to measure the RSS vector at every single location owing to time and cost constraints. Therefore, certain reference points must be properly selected in the positioning environment. By collecting the intensity data of the WiFi signals emitted from different APs at these reference points, we can construct the RSS vector of the reference point with the components of the vector that follow the same order of the AP sequence. These RSS vectors, along with the coordinates of the reference points, can be stored together in the database. Subsequently, an indoor fingerprint database (radio map) is constructed.

Step 2: By comparing and matching the RSS vector of a target location with the data in the fingerprint database, we can determine the position of the target location. The fingerprint-based positioning method does not require an LoS distance measurement from the AP. Therefore, this method can be applied in a complex indoor environment.

4 | WIFI FINGERPRINT-BASED POSITIONING ALGORITHM

The fingerprint-based positioning algorithm can provide positioning without the requirement of an additional measurement of the transformation. It utilizes the existing facility and does not rely on the modification or movement of the hardware. Therefore, maintaining and updating the algorithm will not have a huge impact on the users. Here, we introduce the machine learning methods commonly used in indoor positioning in recent years.

4.1 | K-nearest neighbour (KNN) positioning algorithm

In the KNN positioning algorithm, the RSS vector of a target location is individually compared with all the fingerprints in the radio map. The degree of matching between the RSS vector and the fingerprint is characterized by the Euclidean distance. The Euclidean distances between the target location and all the fingerprints in the ratio map are ranked in the ascending order. Subsequently, the average of the coordinates of the K ($K > 2$) reference points with the smallest distance is taken as the position of the target location [24]. The fingerprint positioning algorithm based on KNN is widely used because of its simple hardware facilities, less complexity, and less computation. When the KNN algorithm is used as the fingerprint matching algorithm, the positioning accuracy is significantly affected by the weight distribution and the K value [25]. Scholars have proposed many methods to improve the positioning accuracy. The study in [26] proposed a weighted K-nearest neighbour method based on the KNN algorithm, which is weighed using the Euclidean distance between the point to be measured and the reference point in the signal space, considering the different effects of each RP on position prediction. The larger the weight of the recent contribution, the weighted mean coordinate is obtained by processing, which improves the positioning accuracy. To address the problem that a fixed K value will introduce a reference point whose actual distance deviates from the to-be-determined point, [27] proposed an adaptive WKNN algorithm, SAWKNN, with a dynamic K value. The value of K is adjusted based on the signal strength to achieve the best accuracy in different locations and different networks. SAWKNN can achieve better localization accuracy than the conventional WKNN.

4.2 | Clustering algorithm

Because of the excessive amount of RSS data, some studies [28, 29] proposed to optimize the management of radio maps using the K-means clustering algorithm before KNN positioning, which effectively shortened the calculation time, to improve the real-time performance of indoor positioning algorithms. The K-means algorithm needs to determine the number of clusters as prior information. The study in [30] proposed affinity propagation clustering, which can automatically generate the number of clusters of the radio map in the iterative process; furthermore, the 'affinity' information will be between the reference points two-way propagation to generate cluster centres. Subsequently, the reference points can be classified into the most suitable clusters. The study in [31] proposed an improved public c-means (IPC) clustering algorithm with high accuracy in indoor environments. The points of the boundary area between two classes are assigned to these classes simultaneously to optimize the clustering effect, thereby improving the localization accuracy of this area. The K-means clustering algorithm can provide good localization performance; however, it cannot remove noisy points. The mean linkage (ML) hierarchical clustering algorithm has been used in some studies [32, 33] in which the mean distance was used as the clustering criterion. The mean distance can better represent the distance between two clusters and overcome the sensitivity issue associated with outliers. The DBSCAN clustering algorithm marks points in low-density regions as outliers. Therefore, it is robust to noise and can deal with noise points efficiently [34]. By combining the DBSCAN clustering algorithm with the K-means clustering algorithm, we can utilize the high localization performance of the K-means algorithm while solving the noise problem.

4.3 | Support vector machine

Support vector machine (SVM) has the advantages of fast convergence speed, strong generalization ability, many adaptation methods, and easy construction, and is often used in classification recognition and model regression. The study in [35] proposed a floor recognition method based on SVM. This method uses the WiFi signal of each floor to identify the floor through SVM. This method uses a linear kernel function for SVM classification and 10-fold cross-validation to train the classifier model. It uses non-linear transformation to transform the input space into a high-dimensional space, and then seeks the best classification in the high-dimensional space. To address the problem of hardware variance and signal fluctuation in WiFi fingerprinting-based localization, [36] proposed an SVM-based solution for room-level prediction, which achieved an accuracy of 98.75%. The positioning performance of SVM is closely related to its parameters. If the parameters are not properly selected, the positioning accuracy will be low and the positioning results will be unreliable. In [37], an improved sparrow search algorithm was proposed to optimize a support vector regression (SVR) machine to obtain an indoor positioning prediction model.

4.4 | Random forest

When the indoor positioning method based on KNN and SVM has a large positioning area and large fingerprint data, the online stage algorithm generally has shortcomings of a large amount of calculation and difficulty in improving the positioning accuracy.

The random forest (RF) method is an ensemble learning algorithm that builds and combines multiple decision trees to achieve highly accurate and stable predictions. RF works very well for large databases. In [38], a hybrid random forest model that takes in signal strength from WiFi devices and uses it to train the model was proposed. The system was tested using WiFi signals obtained from seven different routers. The highest accuracy of 98.3% is achieved using k-fold cross-validation of 3 and mtry = 2. The study in [39] proposed an RF-based wireless indoor positioning system in which RSS readings are taken with four different antenna orientations at each reference point to improve positioning accuracy and mitigate the effect of radio irregularity. The system achieved a precision of 1.69 m.

(4)

4.5 | Artificial neural

When considering a typical location, fingerprint positioning algorithms are essentially machine learning methods based on the shallow structures of non-linear feature transformation, their modelling and expression capabilities are limited, and their structure limits the expression of indoor RSS data features. However, several difficulties are encountered when using complex data. Researchers have attempted to apply deep learning to indoor positioning to achieve improved positioning results. A deep neural network uses the RSS values of all AP nodes at each sampling point in the fingerprint database and the location coordinates of the corresponding sampling points as the training data to obtain the mapping relationship between the RSSI and location coordinates. Furthermore, during online positioning, a set of actual measured RSSs for mapping relationships is input to predict results. It has a strong mapping capability for non-linear features and high flexibility. Moreover, RSS fingerprint data shows good performance during training and learning for indoor positioning. The study in [40] proposed a new deep neural network model. The algorithm attempts to use stacked autoencoders (SAE) to reduce the dimension of the feature space, and expresses the floor classification problem as a multi-label classification problem. Extreme learning machine has the characteristics of fast learning speed and good generalization performance. The study in [42] proposed a floor positioning algorithm based on extreme learning machine. Two layers of ELM-SAE are used for feature extraction, and one layer is used for ELM feature classification. In addition to maintaining its core advantages, this hierarchical learning structure further improves the learning efficiency of the original ELM. The performance of DNN-based methods is still limited by the adequacy of the input training data. Since DNN is fully connected, its computational complexity is directly related to the depth of the neural network (i.e. the number of layers), which directly affects the accuracy of the positioning results. In [43], a localization model combining an SAE with a one-dimensional convolutional neural network (CNN) is proposed. The algorithm uses an SAE to accurately extract key features from sparse WiFi fingerprints, and a 1D-CNN layer to train the classifier model. Inspired by the most advanced image classification methods, the study in [44] proposed a CNN for floor positioning. The WiFi signal is converted into a fingerprint image, and the CNN is used to classify the location of the fingerprint image. This method can reduce sensitivity to received signal fluctuations. Compared with conventional neural networks, the numbers of learning parameters and network complexity are less.

5 | CONCLUSION

The demand for location-based services (LBS) drives the development of indoor positioning technology. A significant effort has been made to develop high precision and instantaneous positioning technology. This study reviewed the main WiFi fingerprint-based algorithms for machine learning. This study is expected to provide an important reference for researchers to realize indoor LBS in people's daily life.

First, we compared the main technologies used in indoor LBS and concluded that the deployment of WiFi in buildings has major advantages. Because most buildings already have WiFi equipment, additional equipment is not required. We then introduced the main WiFi-based indoor technologies, focusing on WiFi fingerprinting algorithms.

Furthermore, WiFi fingerprint-based indoor localization methods are still affected by certain factors, including strong dependence on the environment (e.g. changes in the indoor environment can have a significant impact on indoor localization results). The strength of a WiFi signal is sensitive to many factors, including stationary objects and moving people. The human body is a key source of interference for wireless LAN signal propagation. Humans must participate in the fingerprint collection process in the offline phase and the instantaneous signal reception process in the online phase [46]. Furthermore, walking is a very common activity in indoor environments. Therefore, the human body must be considered in both the initial testing and practical application. This is a limitation of the current study.

CONFLICT OF INTEREST

Authors declare that there is no conflict of Interest.

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DATA AVAILABILITY STATEMENT

Not applicable. Overview of WiFi fingerprinting-based indoor positioning.

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